Predictive Maintenance for Aircraft Engines: A Machine Learning Approach to Remaining Useful Life (RUL) Prediction

DSCI-441 M2 Final Project Report

Abstract

This project develops a supervised machine learning framework to estimate the Remaining Useful Life (RUL) of aircraft turbofan engines from multivariate sensor measurements. The project uses the NASA C-MAPSS FD001 dataset, where each engine is represented by cycle-by-cycle operational settings and sensor readings. RUL labels are constructed from the training trajectories and official RUL labels are used for the test engines. Three regression models were compared: Ridge Regression, Random Forest, and Gradient Boosting. The Random Forest model achieved the best test performance with an RMSE of 18.20 cycles and MAE of 13.18 cycles on capped RUL labels. A Streamlit application was created to demonstrate interactive RUL prediction for uploaded or manually entered engine measurements.

1. Problem Description and Motivation

Aircraft engines operate under safety-critical conditions, and unexpected engine degradation can lead to costly delays, unscheduled maintenance, and safety risks. Traditional maintenance schedules can be conservative and may replace components before their useful life is exhausted. Predictive maintenance uses sensor data and machine learning to estimate the future health of equipment. In this project, the task is to predict the remaining number of operating cycles before failure, known as Remaining Useful Life or RUL.

2. Dataset Description

The project uses the FD001 subset of the NASA C-MAPSS turbofan engine degradation simulation data. The training file contains complete run-to-failure trajectories for 100 engines. The test file contains partial trajectories for 100 engines, and the RUL file gives the true remaining life after the final observed test cycle. Each record includes an engine ID, cycle number, 3 operational settings, and 21 sensor values. The FD001 subset represents one operating condition and one primary fault mode, making it appropriate for a focused RUL regression project.

3. Preprocessing and Feature Engineering

The training target was computed as max engine cycle minus current cycle. A capped RUL target of 130 cycles was used, which is common in turbofan prognostics because early-life measurements often show weak degradation patterns. Constant and low-information columns were removed, including setting_3 and sensors 1, 5, 6, 10, 16, 18, and 19. The final feature set retained operational settings and sensor channels with useful variation for degradation prediction.

4. Model Selection

Three regression models were trained and compared. Ridge Regression served as a linear baseline. Random Forest was used because it can model nonlinear sensor interactions and is robust to noisy measurements. Gradient Boosting was used as an additional ensemble method that builds an additive sequence of weak learners. The final Streamlit app uses the best model selected by lowest test RMSE.

5. Evaluation Results

Model	RMSE	MAE	R2
Random Forest	18.20	13.18	0.800
Gradient Boosting	18.45	12.96	0.794
Ridge Regression	21.23	16.93	0.728

Random Forest achieved the lowest RMSE and was selected as the deployment model. The result indicates that nonlinear ensemble learning is more effective than a purely linear baseline for this sensor-based RUL task.

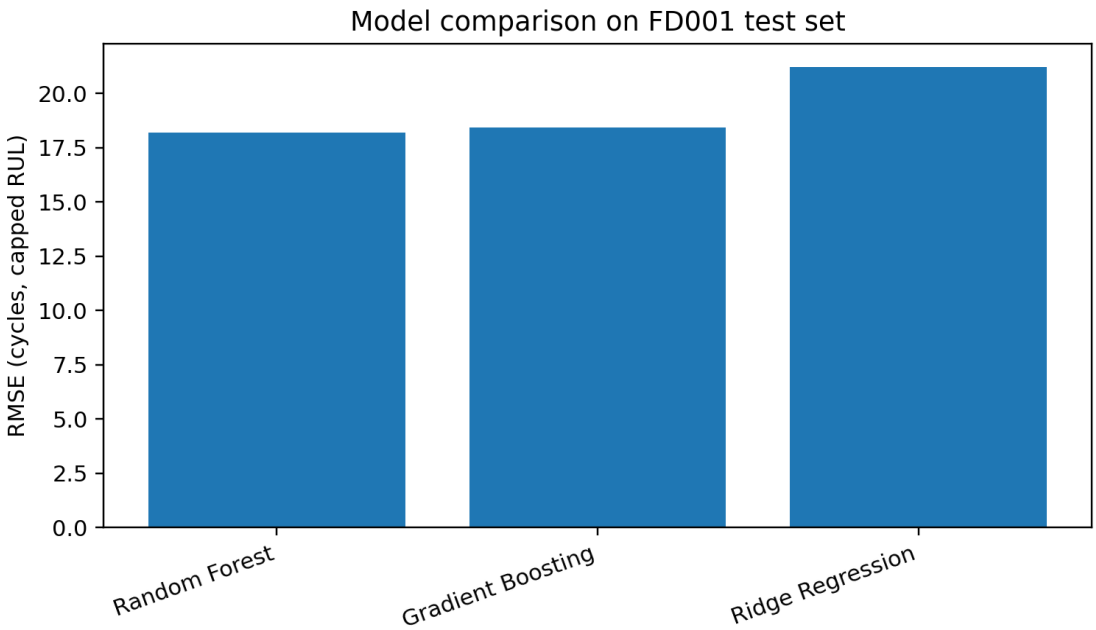

Figure 1. Model comparison by RMSE.

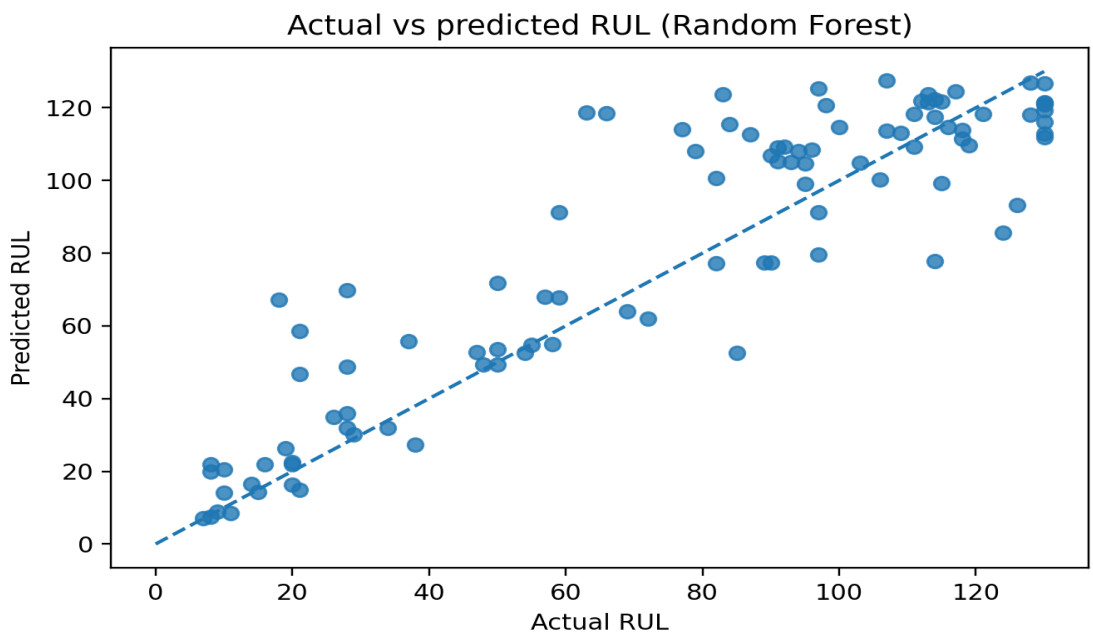

Figure 2. Actual vs predicted RUL for the selected model.

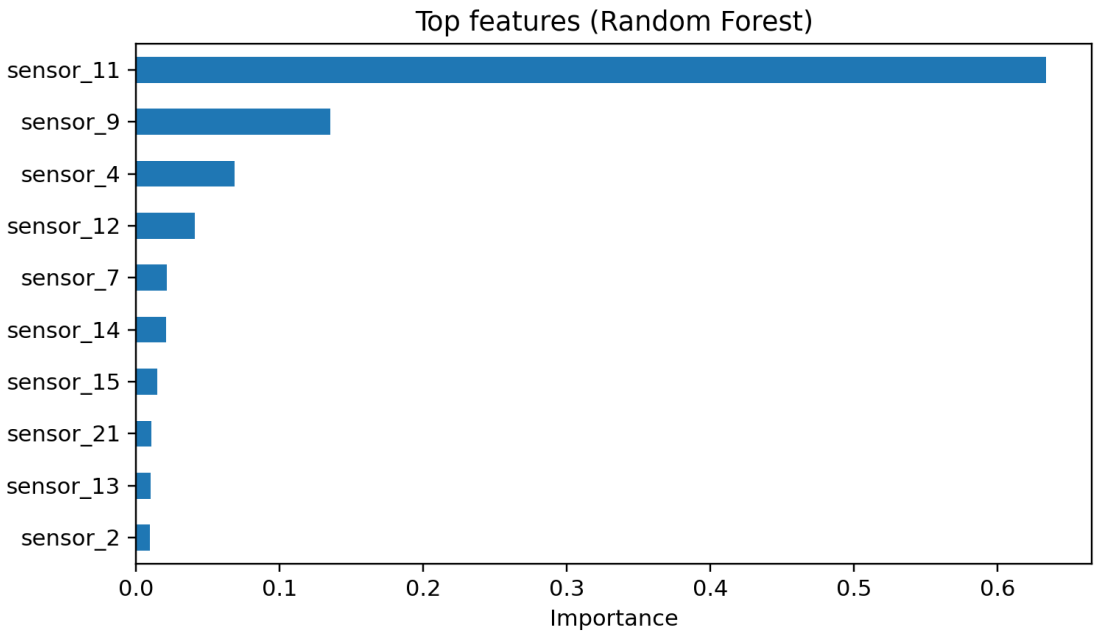

Figure 3. Top feature importances for the selected model.

6. Web Application

A Streamlit web application was implemented in app.py. The app loads the saved Random Forest model and allows two modes of interaction: CSV upload for engine records or manual entry of one current engine observation. The output is a predicted RUL value and a simple maintenance priority message.

7. Conclusion and Future Work

The project demonstrates that machine learning can estimate aircraft engine RUL from multivariate sensor data with practical accuracy. The Random Forest model produced the best overall RMSE among the tested models. Future extensions could include sequence models such as LSTM or temporal convolutional networks, uncertainty intervals for safety-critical decision making, and deployment against streaming engine telemetry.

References

Saxena, A., Goebel, K., Simon, D., & Eklund, N. (2008). Damage propagation modeling for aircraft engine run-to-failure simulation. International Conference on Prognostics and Health Management.

NASA Prognostics Center of Excellence. C-MAPSS Turbofan Engine Degradation Simulation Data.